

Anova Activity #2

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Due Friday, Feb 6, 2026, by 11:59pm

Directions: Complete the questions below. You are encouraged to comment the code

More practice with the ‘InsectSpray Data’

```
data(InsectSprays) #This data set is in the core R package
```

Fitting the Reference Coded Model

```
lm_spray <- lm(count ~ spray, data = InsectSprays)
summary(lm_spray)$coefficients
```

```
##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  14.500000    1.132156  12.8074279 1.470512e-19
## sprayB       0.8333333    1.601110   0.5204724 6.044761e-01
## sprayC      -12.4166667    1.601110  -7.7550382 7.266893e-11
## sprayD      -9.5833333    1.601110  -5.9854322 9.816910e-08
## sprayE     -11.0000000    1.601110  -6.8702352 2.753922e-09
## sprayF       2.1666667    1.601110   1.3532281 1.805998e-01
```

```
favstats(count ~ spray, data = InsectSprays)
```

```
##  spray min    Q1 median    Q3 max    mean    sd  n missing
## 1    A   7 11.50   14.0 17.75  23 14.500000 4.719399 12      0
## 2    B   7 12.50   16.5 17.50  21 15.333333 4.271115 12      0
## 3    C   0  1.00    1.5  3.00   7  2.083333 1.975225 12      0
## 4    D   2  3.75    5.0  5.00  12  4.916667 2.503028 12      0
## 5    E   1  2.75    3.0  5.00   6  3.500000 1.732051 12      0
## 6    F   9 12.50   15.0 22.50  26 16.666667 6.213378 12      0
```

Review: Using the reference-coded output above, what is the estimated mean number of insects per hectare for each insecticide? What is the formula for $\hat{\mu}_j$ for all j from the reference-coded model? What are the values of each estimate?

Reminder: you can create an “alpha-hat” using $\hat{\alpha}$ and a “tau-hat-j” using $\hat{\tau}_j$. Use the format below to report your results for each insecticide.

- **Spray A:** $\hat{\mu}_A = 14.5000000 + 14.5000000 = 29$ (insects per hectare)
- **Spray B:** $\hat{\mu}_B = 14.5000000 + 0.8333333 = 15.33$ (insects per hectare)
- **Spray C:** $\hat{\mu}_C = 14.5000000 - 12.4166667 = 2.0833333$ (insects per hectare)
- **Spray D:** $\hat{\mu}_D = 14.5000000 - 9.5833333 = 4.916667$ (insects per hectare)
- **Spray E:** $\hat{\mu}_E = 14.5000000 - 11.0000000 = 3.5$ (insects per hectare)
- **Spray F:** $\hat{\mu}_F = 14.5000000 + 2.1666667 = 16.666667$ (insects per hectare)

Are these the same as the sample means for each insecticide?

- They are the same as the sample means for each insecticide except for Spray A

One-Way ANOVA Sums of Squares, Mean Squares, and F Test

- Now that we know the two ways of parameterizing the alternative “separate means” model and getting how to get estimates from the output, we can learn how to assess the evidence related to whether the observed differences in means among the groups is “statistically large”. (Remember “statistically large” does not equal “practically large!”)
- We graphically examined the components of the F-test and how they relate the variability *explained* by the alternative model to the variability left *unexplained* by the alternative model. Thus, we developed an understanding of the ANOVA F-Statistic, which is the statistic that allows the comparison of relative amounts of variation.
- The F-statistic, if certain assumptions are met, follows the F-distribution. The F-distribution is a right-skewed distribution whose shape is defined by the *numerator degrees of freedom* ($J - 1$) and *denominator degrees of freedom* ($N - J$). Since the F-distribution is defined on positive numbers, we always use the right tailed area for p-values. The components of an ANOVA F-Test can be summarized into the following table. Recall that N is the total number of observations and J is the number of groups.

General One-Way ANOVA Table

Source	DF	Sums of Squares	Mean Squares	F-ratio	P-value
Variable A	$J - 1$	SS_A	$MS_A = SS_A / (J - 1)$	$F = MS_A / MS_E$	Right tail of $F(J - 1, N - J)$
Residuals	$N - J$	SS_E	$MS_E = SS_E / (N - J)$		
Total	$N - 1$	SS_{Total}			

Here is a partially filled-in ANOVA table for the InsectSprays dataset including all sprays (A,B,C,D,E,F). **On the paper provided, fill in the missing values based on the general format of the table and show work for how you calculated each value. You will turn in a picture of your work associated with this question. Do not enter values into the table below!**

Source	DF	Sum of Squares	Mean Squares	F-ratio	P-Value
spray		2668.8			

Source	DF	Sum of Squares	Mean Squares	F-ratio	P-Value
Residuals					

Total 3684

- Calculate the p-value for the ANOVA F-test (fill in the code below to get the p-value):

```
pf(34.7, df1=5, df2=66, lower.tail=F)
```

```
## [1] 3.187508e-17
```

- R can also generate the ANOVA table by applying the `anova` function to the linear model. It doesn't have the *total* row as the general form of the table does. Do the values you filled in match? Run the code in this example to check! **Revise your work until you get the correct values!**

```
anova(lm_spray)
```

- Write a conclusion including a summary of statistical findings to address the question, "*Are there any differences in the true mean number of insects per hectare counted among the different insecticide treatments?*" Be sure to include a parenthetical summary of statistical findings in your conclusion (i.e., (Observed test stat, its assumed distribution under H_0 , associated p-value)).
- **There is strong evidence of a difference in the mean number of insects per hectare among differing insecticide treatments. The observed F-stat is 34.7, which under the null hypothesis follows an F-distribution with 5 and 66 degrees of freedom. The p value is less than 0.0001, indicating that at any conventional significance level we reject the null hypothesis. This suggests that one insecticide has a mean insect count significantly differing from the others**